Table 5: Learned parameters for each Smokers and friends experiment, to contrast ablation of logical connective weights, and show the effect of a larger simultaneous axiom set on bounds relaxation.

Experiment	Axioms	Axiom bounds	Initial fact bounds	Connective weights
P_1^5	5	✓	✓	✓
P_2^5	5	✓	✓	
P_1^8	8	✓	✓	✓
P_2^8	8	✓	✓	

After each update a new training epoch is initialized with the updated bounds, followed by the Upward-Downward inference algorithm run until no new inferences are made. Backpropagation occurs over all of these inference passes for a given training epoch to provide a gradient trace to the initial computational graph leaves that include the adjusted fact/axiom bounds and logical connective weights. The learning settings in Table 6 are all the same for the various experiments, and a relatively small number of epochs are required.

Table 6: Learning settings for $P_1^5, P_2^5, P_1^8, P_2^8$, with a decreasing learning rate schedule over 100 epochs from 0.1→0. Gradient clipping for truth bound and logical connective weights ensure granular updates, and a minimum logical connective weight is kept to avoid division-by-zero during inference.

Epochs	**Learning rate** (start)	**Learning rate** (end)	**Weight** (min)	**Gradient clipping**
100	0.1	0	0.01	[-0.1, 0.1]

Experimental results. Overall results in Tables 7 and 8 show learnt initial fact and axiom truth value bounds for the different experiments, also showing additional loss values not given in the main paper. The LTN degree of satisfiability could possibly be compared to the mean of the LNN truth value bounds, although some comparative deviances are noted especially for $\neg S(x) \vee \neg F(x, y) \vee S(y)$ where LTN reports high satisfiability of 0.96 while LNN records bounds of [65, 100]. This is possibly due to LTN not completely inferring friendship symmetry, whereas LNN applies friendship symmetry which is then followed by transitivity of smoking amongst friends that are untrue in some cases.

The MLN formula weights are not directly comparable to truth values, and requires conversion to probabilities, although this conversion would be dependent on the data specifically. MLN axiom weight 6.88 for $\exists y F(x, y)$ states that a world with n friendless people is $e^{6.88n}$ times less probable than a world where all have friends, other things being equal [15], but LNN sets full truth as it does not conflict. Still, a relative magnitude ranking can be compared, where we see that the last two axioms have relatively high weight or log-probability of being true for the data, in spite of the observation of many contradictions attributed to these axioms. This could possibly due to incorrect closed-world assumptions made about participating predicates.

The primary observation is that LNN can perform gradient descent with the system loss to remove contradiction by relaxing facts and axioms, carefully balancing impact on factual correctness and the usefulness (tightness) of inferences made. If facts deviate too much from groundtruth then factual incorrectness arises, and when an axiom truth is relaxed too much, then less tight and more unknown inferences will be made. Learning of neuron weights can possibly reduce bounds relaxation in P_1^8 when comparing to P_2^8 for the last two high-conflict axioms.

Clamped gradients. We perform an experiment to measure the effect of available gradients across the clamped domain of Łukasiewicz for the 8 axiom set where we update all parameters, namely P_1^8. The results in Figure 7 shows graphs for the system loss $(1 + \texttt{contradiction})/(1 + \texttt{factalign} + \texttt{tightbounds})$ and its denominator components. Gradient supplantation, introduced in Section G.3.4, is used here to introduce arbitrary gradients in the clamped domain. A comparison is made against the baseline of conventional Łukasiewicz with no gradients in the clamped domain.

We note a "flip-flop" learning convergence pattern for the loss in Figure 7, where slight contradictions register in alternate epochs which are subsequently corrected. This is possibly due to the optimum closely bordering a logically inconsistent state, so as not to relax bounds and axioms too much. Gradients detach for corrected contradictions, which mean that only alternate epochs show non-zero

Table 7: Inferred fact bounds ($[L, U]$ as %) for LTN $\mathcal{K}_{\text{exp2}}$ (cmp. [19] Table 1) with learnt initial facts. A single number indicates matching bounds $L = U$. The purple shaded regions have groundtruth facts provided, which are adjusted through learning, followed by a final inference to convergence to produce the bounds below.

P_1^5			Friends							
	Smokes	Cancer	a	b	c	d	e	f	g	h
a	**[63, 100]**	**[98, 100]**	[0, 17]	**[66, 100]**	[0, 23]	[0, 23]	**[99, 100]**	**[99, 100]**	**[99, 100]**	[0, 23]
b	[0, 39]	0	**[63, 100]**	[0, 17]	**[99, 100]**	[0, 23]	[0, 23]	[0, 23]	[0, 23]	[0, 23]
c	[0, 20]	0	[0, 27]	**[96, 100]**	[0, 17]	**[99, 100]**	[0, 23]	[0, 23]	[0, 23]	[0, 23]
d	[0, 20]	0	[0, 27]	[0, 27]	**[96, 100]**	[0, 17]	[0, 23]	[0, 23]	[0, 23]	[0, 23]
e	**[80, 100]**	**100**	**[96, 100]**	[0, 27]	[0, 27]	[0, 27]	[0, 17]	**[99, 100]**	[0, 23]	[0, 23]
f	**[58, 82]**	[16, 42]	**[96, 100]**	[0, 27]	[0, 27]	[0, 27]	**[96, 100]**	[0, 17]	[0, 23]	[0, 23]
g	**[58, 82]**	[16, 42]	**[96, 100]**	[0, 27]	[0, 27]	[0, 27]	[0, 27]	[0, 27]	[0, 17]	**[70, 100]**
h	[0, 37]	0	[0, 27]	[0, 27]	[0, 27]	[0, 27]	[0, 27]	[0, 27]	**[67, 100]**	[0, 17]

P_2^5			Friends							
	Smokes	Cancer	a	b	c	d	e	f	g	h
a	**[63, 100]**	**[97, 100]**	[0, 17]	**[66, 100]**	[0, 22]	[0, 22]	**99**	**99**	**99**	[0, 22]
b	[0, 40]	0	**[63, 100]**	[0, 17]	**99**	[0, 22]	[0, 22]	[0, 22]	[0, 22]	[0, 22]
c	[0, 20]	0	[0, 24]	**[96, 100]**	[0, 17]	**99**	[0, 22]	[0, 22]	[0, 22]	[0, 22]
d	[0, 20]	0	[0, 24]	[0, 24]	**[96, 100]**	[0, 17]	[0, 22]	[0, 22]	[0, 22]	[0, 22]
e	**[80, 100]**	**100**	**[96, 100]**	[0, 24]	[0, 24]	[0, 24]	[0, 17]	**99**	[0, 22]	[0, 22]
f	**[59, 83]**	[17, 41]	**[96, 100]**	[0, 24]	[0, 24]	[0, 24]	**[96, 100]**	[0, 17]	[0, 22]	[0, 22]
g	**[59, 83]**	[17, 41]	**[96, 100]**	[0, 24]	[0, 24]	[0, 24]	[0, 24]	[0, 24]	[0, 17]	**[70, 100]**
h	[0, 36]	0	[0, 24]	[0, 24]	[0, 24]	[0, 24]	[0, 24]	[0, 24]	**[67, 100]**	[0, 17]

P_1^8			Friends							
	Smokes	Cancer	a	b	c	d	e	f	g	h
a	**[62, 100]**	**[77, 100]**	[0, 44]	**[71, 100]**	[0, 39]	[0, 39]	**[78, 100]**	**[78, 100]**	**[75, 100]**	[0, 39]
b	[0, 50]	[0, 25]	**[23, 99]**	[0, 44]	**[75, 100]**	[0, 41]	[0, 39]	[0, 39]	[0, 39]	[0, 41]
c	[0, 47]	[0, 25]	[0, 88]	**[27, 100]**	[0, 44]	**[76, 100]**	[0, 39]	[0, 39]	[0, 39]	[0, 41]
d	[0, 47]	[0, 24]	[0, 88]	[0, 90]	**[28, 99]**	[0, 44]	[0, 39]	[0, 39]	[0, 39]	[0, 41]
e	**[65, 100]**	**[80, 100]**	**[30, 98]**	[0, 88]	[0, 88]	[0, 88]	[0, 44]	**[80, 100]**	[0, 41]	[0, 39]
f	**[50, 100]**	[0, 50]	**[30, 97]**	[0, 88]	[0, 88]	[0, 88]	**[31, 97]**	[0, 44]	[0, 41]	[0, 39]
g	**[50, 100]**	[0, 50]	**[27, 99]**	[0, 88]	[0, 88]	[0, 88]	[0, 90]	[0, 90]	[0, 44]	**[71, 100]**
h	[0, 50]	[0, 25]	[0, 88]	[0, 90]	[0, 90]	[0, 89]	[0, 88]	[0, 88]	**[24, 98]**	[0, 44]

P_2^8			Friends							
	Smokes	Cancer	a	b	c	d	e	f	g	h
a	**[62, 100]**	**[80, 100]**	[0, 19]	**[67, 100]**	[0, 20]	[0, 20]	**[78, 100]**	**[79, 100]**	**[77, 100]**	[0, 20]
b	[0, 35]	[0, 24]	**[49, 100]**	[0, 19]	**[75, 100]**	[0, 22]	[0, 20]	[0, 20]	[0, 20]	[0, 22]
c	[0, 30]	[0, 23]	[0, 39]	**[57, 100]**	[0, 19]	**[77, 100]**	[0, 20]	[0, 20]	[0, 20]	[0, 22]
d	[0, 29]	[0, 21]	[0, 39]	[0, 41]	**[58, 100]**	[0, 19]	[0, 20]	[0, 20]	[0, 20]	[0, 22]
e	**[67, 100]**	**[80, 100]**	**[59, 100]**	[0, 39]	[0, 39]	[0, 39]	[0, 19]	**[79, 100]**	[0, 23]	[0, 20]
f	**[59, 81]**	[18, 41]	**[60, 100]**	[0, 39]	[0, 39]	[0, 39]	**[60, 100]**	[0, 19]	[0, 23]	[0, 20]
g	**[59, 81]**	[19, 41]	**[58, 100]**	[0, 39]	[0, 39]	[0, 39]	[0, 41]	[0, 41]	[0, 19]	**[71, 100]**
h	[0, 35]	[0, 23]	[0, 39]	[0, 41]	[0, 41]	[0, 41]	[0, 39]	[0, 39]	**[52, 98]**	[0, 19]

a-h *Initialized facts*

Table 8: Learnt LNN neuron weights (from P_1^8) and axiom lower bounds (as %) for LTN experiment $\mathcal{K}_{\text{exp2}}$ (universe: a-h) [19], compares LTN degree of satisfiability (as % for 5 axioms) to LNN P_1^5, P_2^5 and repeats in P_1^8, P_2^8 for 8 axioms including 3 induced by MLN [15], with corresponding MLN log-probability weights, followed by axiom-wise contradiction counts. Loss function $(1+\texttt{contradiction})/(1+\texttt{factalign}+\texttt{tightbounds})$ (normalized) component values after training show complete removal of contradictions by relaxing facts and inferences. Gradient descent ($\alpha = 1, \beta = 1, w_{\max} = 1$ with weight normalization and gradient-transparent clamping) adjusts operand weights (P_1), initial axiom/fact bounds (P_1, P_2). Every training epoch performs inference initialized with updated bounds until convergence after the parameter update.

		LNN $[L, U]$	*100 epochs, lr: 0.1→0*				
Smokers and friends [$\mathcal{K}_{\text{exp2}}$ (a-h)]	**LTN**	P_1^5	P_2^5	P_1^8	P_2^8	**MLN**	$L{>}U$
$\exists y F(x,y)$	100	100	100	100	100	6.88	0
$\neg F(x,x)$	98	[83, 98]	[83, 98]	[56, 98]	[80, 98]	0.26	0
$\neg F(x,y)^{0.96} \vee F(y,x)^1$	90	[96, 97]	[97, 100]	[51, 95]	[82, 97]	-	0
$\neg S(x)^{0.98} \vee \neg F(x,y)^1 \vee S(y)^{0.97}$	96	[65, 100]	[65, 100]	[65, 100]	[66, 100]	3.53	2
$\neg S(x)^1 \vee C(x)^{0.98}$	77	[57, 100]	[58, 100]	[50, 100]	[60, 100]	-1.35	2
$\neg F(x,y)^{0.97} \vee \neg S(y)^1 \vee F(y,x)^{0.96} \vee S(x)^1$				100	100	6.87	0
$\neg F(w,x)^1 \vee \neg F(w,y)^1 \vee \neg F(z,x)^1 \vee C(z)^{0.97}$				[73, 100]	[70, 100]	4.33	51
$\neg F(w,x)^1 \vee \neg F(y,w)^1 \vee \neg F(z,y)^1 \vee \neg S(y)^{0.99}$				[80, 100]	[77, 100]	9.68	66
Contradiction (remaining)		0	0	0	0		*121*
Factual $E_i[\lvert L_i' - L_i\rvert + \lvert U_i' - U_i\rvert]$ (start: 0.64)		0.42	0.43	0.27	0.37		
Bound tightness $E_i[\exp(L_i - U_i)]$ (start: 1)		0.88	0.89	0.9	0.97		
Loss (start)		1.897	1.897	46.276	46.276		
Loss (end)		0.435	0.432	0.461	0.429		

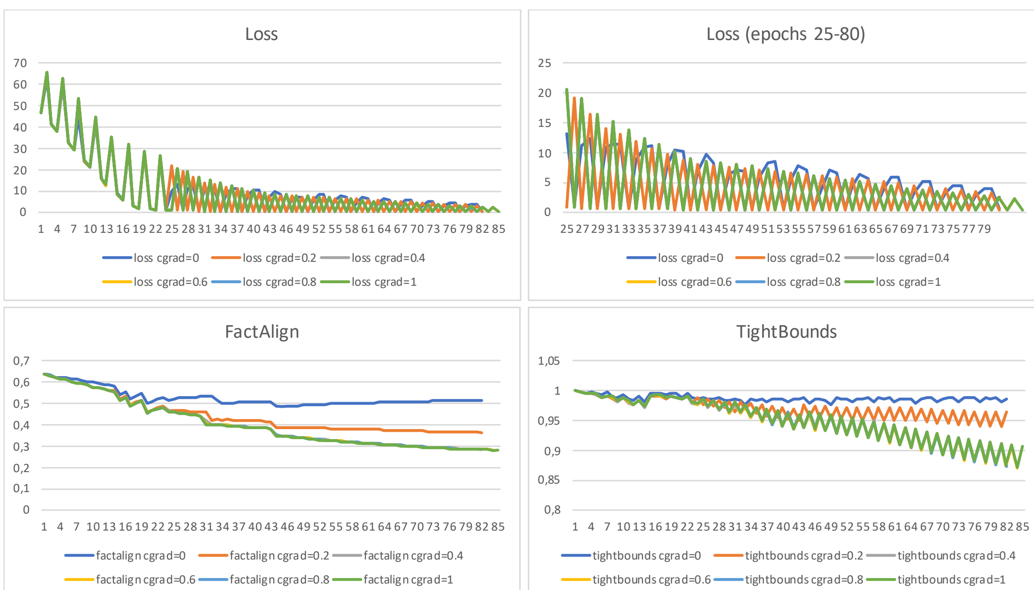

Figure 7: Gradient supplantation $x(0 \le x) + c_{\text{grad}}(x - x^*)(x < 0)$ (please see Section G) with arbitrary Łukasiewicz conjunction `False` gradients $c_{\text{grad}} \in \{0, 0.2, 0.4, 0.6, 0.8, 1\}$ are compared in terms of the system loss $(1+\texttt{contradiction})/(1+\texttt{factalign}+\texttt{tightbounds})$ and its denominator components for P_1^8. A conventional Łukasiewicz conjunction `False` gradient $c_{\text{grad}} = 0$ changes fewer parameters, but is still able to remove contradictions for this CNF system although at a slower convergence, yet it exhibits smaller deviance from initial facts and produces more useful inferences. Łukasiewicz conjunctions with non-zero `False` gradients appear to contribute to converging slightly faster (at the top of the "flip-flop"), as it possibly has more parameters with gradients available, although the downside is larger relaxation of facts and axioms.